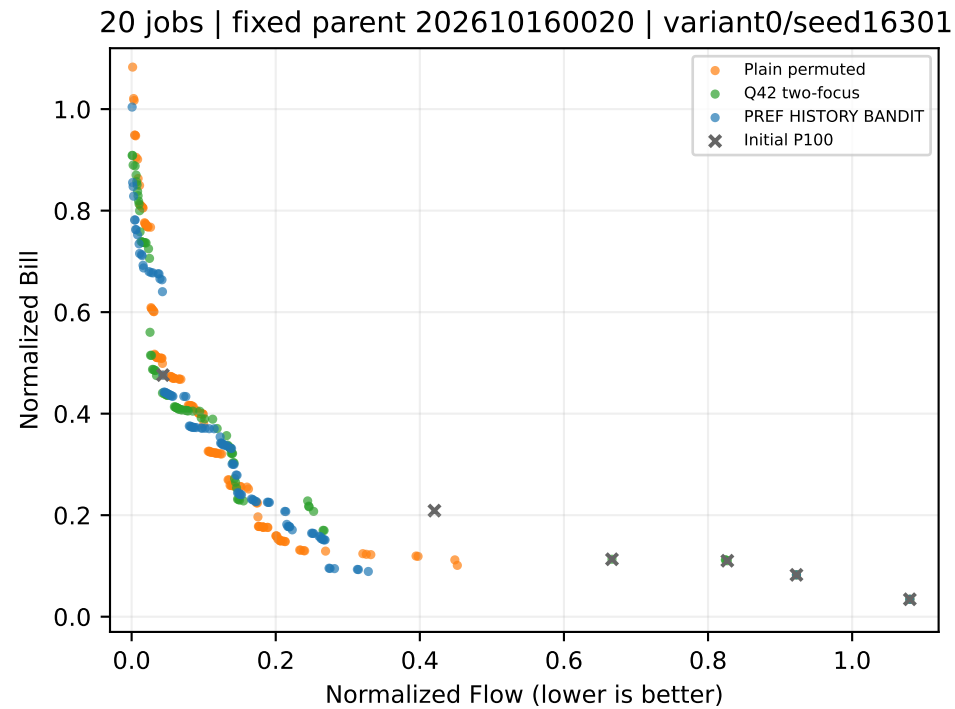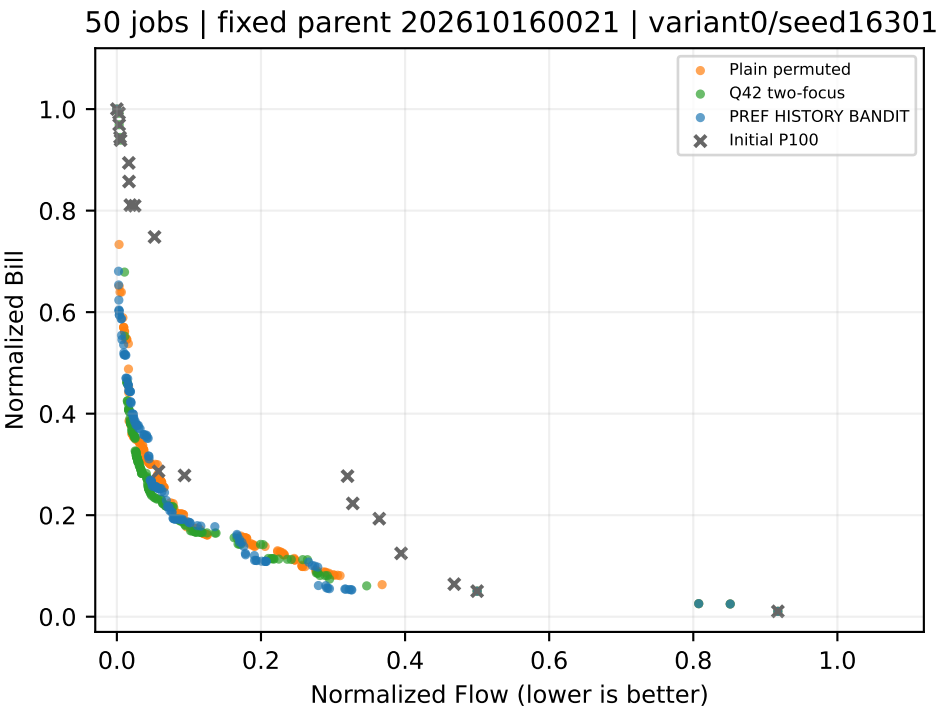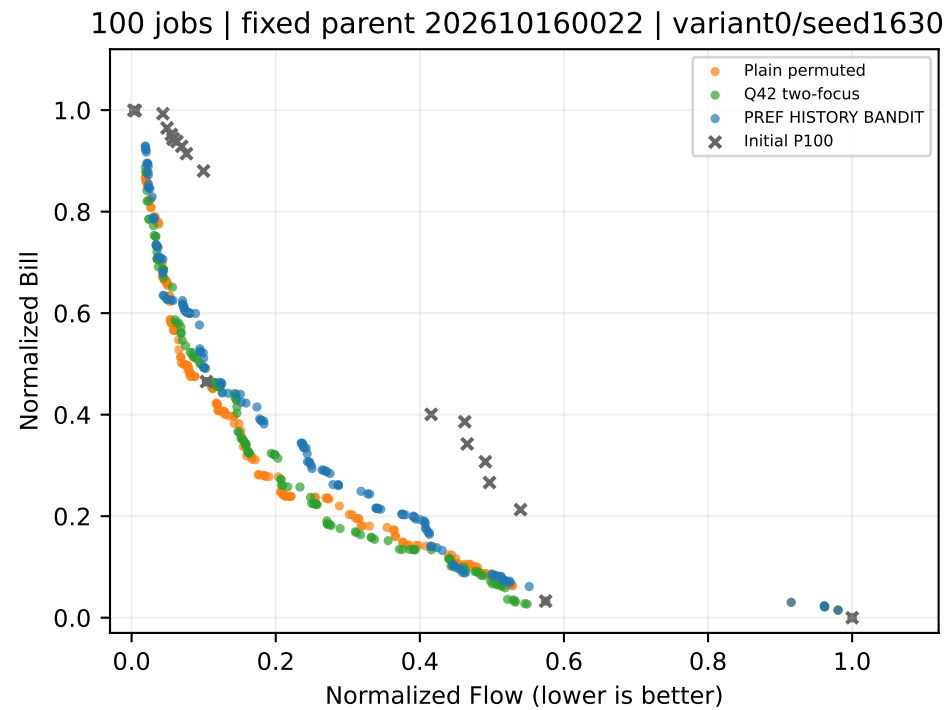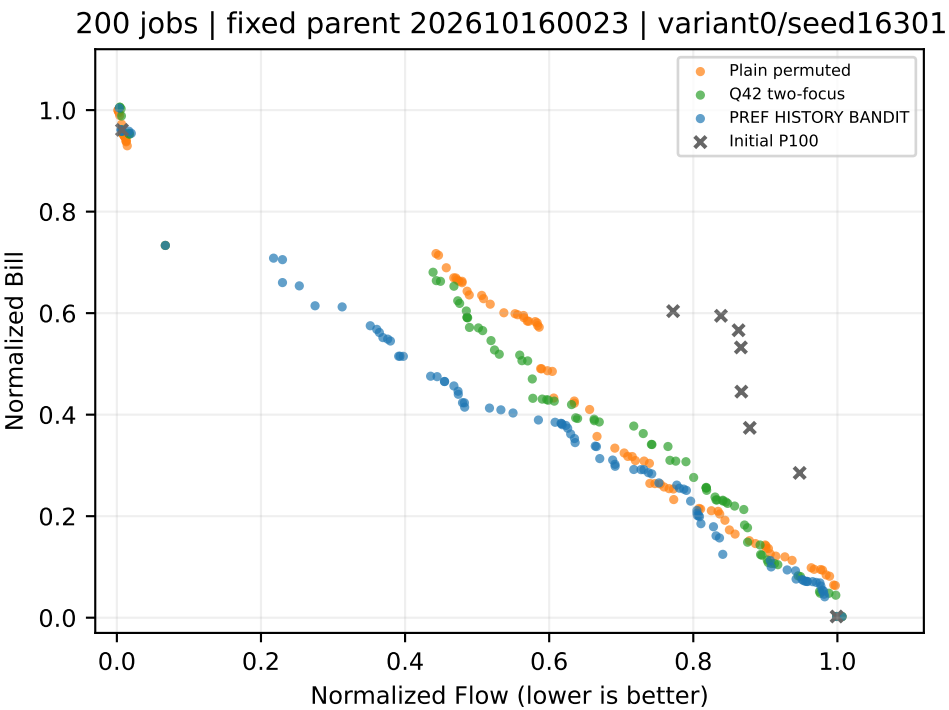PREF HISTORY BANDIT: fixed cases, no best-case selection; empirical union normalization